CONCRETE WORKS

MES-ST-SPC-0005

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for construction of concrete works for all Ma'aden projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum standards for furnishing and installation of all materials, forming, placing, consolidating, finishing, jointing, curing, testing and other related requirements for plain and reinforced concrete.

1.3 Conflicts and Deviations

Any deviation from any of the requirements stipulated in this document shall go thru a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products, materials, and/ or items provided in accordance with contract documents
Supplier	Party responsible for supplying concrete materials in accordance with contract documents

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health,

Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0004	Premixed Concrete.
MD-101-SMEM-EG-ST-SPC-0006	Steel Reinforcing Bars and Wire Fabric.
MD-101-SMEM-EG-ST-SPC-0008	Protective Coating for Concrete.

3.3 Industry Codes and Standards

All buildings, equipment and systems shall conform to the latest adopted revision of the Codes and standards in the following paragraphs. In case of conflict in general the most stringent requirements shall govern.

3.3.1 *Local Codes and Standards (for Saudi Arabia)*

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

SBC 304	Structural – Structural Concrete
SASO/SSA142	Physical or Mechanical Methods of Testing of Portland Cement
SASO/SSA143	Portland Cement Ordinary and Rapid Hardening

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.3.2 *International Codes and Standards*

3.2.2.1. American Concrete Institute (ACI) (use metric equivalent when applicable to Project Site)

ACI SPEC-117	Specification for Tolerances for Concrete Construction and Materials
ACI PRC-201.2	Durable Concrete - Guide
ACI PRC-214	Guide to Evaluation of Strength Test Results of Concrete
ACI PRC-224.1	Causes, Evaluation, and Repair of Cracks in Concrete Structures
ACI PRC-224.3	Joints in Concrete Construction
ACI PRC-237	Self-Consolidating Concrete
ACI SPEC-301	Specifications for Concrete Construction
ACI PRC-302.1	Guide for Concrete Floor and Slab Construction
ACI PRC-304	Guide for Measuring, Mixing, Transporting and Placing Concrete
ACI PRC-304.2	Guide to Placing Concrete by Pumping Methods
ACI SPEC-305.1	Guide for Hot Weather Concreting
ACI SPEC-306.1	Standard Specification for Cold Weather Concreting
ACI SPEC-308.1	External Curing of Cast-in- Place Concrete- Specification
ACI PRC-309	Guide for Consolidation of Concrete
ACI PRC-315	Guide to Presenting Reinforcing Steel Design Details
ACI CODE-318	Building Code Requirements for Structural Concrete
ACI CODE-440.11	Building Code Requirements for Structural Concrete Reinforced with Glass Fiber-Reinforced Polymer (GFRP) Bars
ACI PRC-347	Guide to Formwork for Concrete
ACI CODE-350	Code Requirements for Environmental Engineering Concrete Structures
ACI SPEC-350.1	Tightness Testing of Environmental Engineering Concrete Structures-Specification.
ACI SPEC-350.5	Specification for Environmental Concrete Structures

ACI PRC-523.2	Guide for Precast Cellular Concrete Floor, Roof and Wall Unit
ACI PRC-533	Guide for Precast Concrete Wall Panels
ACI ITG-7	Specification for Tolerances for Precast Concrete
ACI MNL-66	ACI Detailing Manual.

3.2.2.2. American Society for Testing and Materials (ASTM)

ASTM A1064/A1064M	Standard Specification for Carbon- Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A615/A615M	Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A706/A706M	Standard Specification for Deformed and Plain Low-Alloy Steel Bars for Concrete Reinforcement
ASTM A775/A775M	Standard Specification for Epoxy-Coated Steel Reinforcing Bars
ASTM A884/A884M	Standard Specification for Epoxy-Coated Steel Wire and Welded Wire Reinforcement
ASTM A853	Standard Specification for Steel Wire, Carbon, for General Use
ASTM C31/C31M	Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C33/C33M	Standard Specification for Concrete Aggregates
ASTM C40/C40M	Standard Test Method for Organic Impurities in Fine Aggregates for Concrete
ASTM C42/C42M	Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
ASTM C88/C88M	Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate

ASTM C94/C94M	Standard Specification for Ready-Mixed Concrete
ASTM C114	Standard Test Methods for Chemical Analysis of Hydraulic Cement
ASTM C117	Standard Test Method for Materials Finer Than 75µm (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C127	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregate
ASTM C128	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Fine Aggregate
ASTM C131/C131M	Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C138/C138M	Standard Test Method for Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete
ASTM C142/C142M	Standard Test Method for Clay Lumps and Friable Particles in Aggregates
ASTM C143/C143M	Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	Standard Specification for Portland Cement
ASTM C171	Standard Specification for Sheet Materials for Curing Concrete
ASTM C39/C39M	Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C172/C172M	Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	Standard Test Method for Air Content of freshly Mixed Concrete by the Volumetric Method
ASTM C191	Standard Test Method for Time of Setting of Hydraulic Cement by Vicat Needle

ASTM C192/C192M	Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory
ASTM C231/C231M	Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C295/C295M	Standard Guide for Petrographic Examination of Aggregates for Concrete
ASTM C494/C494M	Standard Specification for Chemical Admixtures for Concrete
ASTM C531	Standard Test Method for Linear Shrinkage and Coefficient of Thermal Expansion of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
ASTM C579	Standard Test Methods for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
ASTM C642	Standard Test Method for Density, Absorption, and Voids in Hardened Concrete
ASTM C920	Standard Specification for Elastomeric Joint Sealants
ASTM C1017/C1017M	Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete
ASTM C1064/C1064M	Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete
ASTM C1077	Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM C1152/C1152M	Standard Test Method for Acid-Soluble Chloride in Mortar and Concrete
ASTM C1240	Standard Specification for Silica Fume Used in Cementitious Mixtures
ASTM C1610/C1610M	Standard Test Method for Static Segregation of Self-Consolidating Concrete Using Column Technique

ASTM C1611/C1611M	Standard Test Method for Slump Flow of Self-Consolidating Concrete
ASTM C1621/C1621M	Standard Test Method for Passing Ability of Self-Consolidating Concrete by J-ring
ASTM C1712	Standard Test Method for Rapid Assessment of Static Segregation Resistance of Self-Consolidating Concrete Using Penetration Test
ASTM D512	Standard Test Methods for Chloride Ion in Water
ASTM D516	Standard Test Method for Sulphate Ion in Water
ASTM D6690	Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavement
ASTM D1751	Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Asphalt Types)
ASTM D2103 REV A	Standard Specification for Polyethylene Film
ASTM D2628	Standard Specification for Preformed Polychloroprene Elastometric Joint Seals for Concrete Pavements
ASTM D5249	Standard Specification for Backer Material for Use with Cold-and Hot-Applied Joint Sealants in Portland Cement Concrete and Asphalt Joints
ASTM D5893/5893M	Standard Specification for Cold-Applied, Single-Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements
ASTM D7957/D7957M	Standard Specification for Solid Round Glass Fiber Reinforced Polymer Bars for Concrete Reinforcement
ASTM E11	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E1745	Standard Specification for Plastic Water Vapor Retarders Used in Contact with Soil or Granular Fill under Concrete Slabs

3.2.2.3. International Standardizations Organization (ISO)

ISO 9001:2015	Quality management system - Requirements

3.2.2.4. Process Industry Practices (PIP)

STS03001	Plain and Reinforced Concrete Specification
STF05121	Anchor fabrication and Installation into Concrete

3.2.2.5. American Association of State Highway and Transportation Officials (AASHTO)

AASHTO GDPS	Guide for Design of Pavement Structures – Rev 4

3.2.2.6. Corps of Engineers (CE)

CE CRD-C572	Handbook for Concrete and Cement Corps of Engineers Specifications for Polyvinylchloride Waterstops.

3.3.3 *Alternative Design Codes*

Alternative codes may be used for concrete works if approved by the Employer.

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. MATERIALS AND CONSTRUCTION

4.1 Concrete

All concrete shall be manufactured in accordance with Ma'aden standard specification, MD-101-SMEM-EG-ST-SPC-0004, Premixed Concrete.

The Contractor shall be responsible for the correct grade of mix being delivered to site and placed in accordance with this specification.

The use of admixtures shall be in accordance with standard specification, MD-101-SMEM-EG-ST-SPC-0004, Premixed Concrete.

All below grade concrete shall be protected by 3 coats of bitumen under polyethylene sheet liner conforming to ASTM D2103, with a minimum thickness of 0. 25mm.The liner shall extend a minimum of 150 mm beyond all concrete foundation edges. In foundation, the underside of concrete foundation, where bitumen and polyethylene cannot be used, shall also be protected against chemical attack. The blinding concrete shall be treated with crystalline waterproofing technology or similar approved, prior to placing the concrete. Also, as an alternative to bitumen/polyethylene

protection, consider using GFRP reinforcing bars, increasing concrete cover by 25mm and/or using concrete with w/c ratio less than 0.4.

4.2 Reinforcement

4.2.1 *General*

Reinforcing bars shall be deformed billet steel conforming to ASTM A615/A615M grade 60 unless noted otherwise on the drawings. Where stated on the drawings or contract documents reinforcement shall be epoxy coated in accordance with ASTM A775/A775M. When epoxy coated reinforcement is specified, all reinforcement within 500 mm of finished grade shall be epoxy coated unless specified otherwise on the drawings or contract documents.

Welded wire fabric shall normally be plain wire fabric and conform to ASTM A1064/A1064M. Where stated on the drawings or contract documents welded wire fabric shall be epoxy coated in accordance with ASTM A884/A884M.

Tie wire shall be black annealed wire, minimum 1.6 mm diameter, and conform to ASTM A853. Epoxy coated wire shall be used where wherever epoxy coated bars are used. Glass Fiber Reinforced Polymer Bars (GFRP) to conform to ASTM D7957/D7957M. Consider using GFRP for seawater structures and non-fire structures.

4.2.2 *Fixing*

The reinforcement shall be fixed in accordance with the requirements of ACI SPEC-301, in the positions shown on the drawings, and in such a manner as to avoid displacement before, during and after concreting. All intersections of reinforcement shall be securely tied by 1.6 mm diameter annealed iron wire, or any other method approved by the Employer.

When fixing reinforcement, the Contractor shall provide all necessary tying wire, chairs and hanger bars to retain all reinforcing bars and fabric in the correct position.

Only concrete or non-metallic supports shall be used between the reinforcement and the finished face of the concrete. Plastic tipped wire bar chairs shall not be used.

The ends of all binding wire shall be turned into the concrete and shall not be left in contact with the shuttering.

Concrete cover to bars shall be in accordance with the requirements of ACI SPEC-301 and as specified on the drawings. The nominated cover shall be to the outermost bar (including ties) of the concrete element.

Reinforcement splices shall be in accordance with the requirements of ACI SPEC-301, and with the details shown on the drawings. In splices, the bars shall be in contact and tied together. Splices shall only be made where indicated on the drawings, or as approved by the Employer.

Immediately prior to each instance of placement of concrete, all loose rust, oil, paint, dried mortar and other foreign substances affecting bond shall be cleaned from all reinforcement within that portion of the works.

The minimum distance between individual bars shall not be less than maximum size of coarse aggregate +5mm, diameter of reinforcement bar and 50mm whichever is greater

4.3 Detailed Earthworks for Foundations

All excavations and backfilling for concrete works shall be carried out in accordance with standard specification, MD-101-SMEM-EG-SCV-SPC-0001, Earthworks.

4.4 Formwork

4.4.1 *General*

All formwork shall be designed and constructed to the requirements and classifications of ACI PRC-347, with prior approval of the Employer.

The finished concrete shall be within the tolerances required by this specification and ACI SPEC-117, and these tolerances shall be generally applicable unless otherwise shown on the drawings or where the general appearance or purpose of the structure is adversely affected.

4.4.2 *Stripping*

The Contractor shall remove all formwork in such a manner as to ensure the complete safety of all personnel engaged in this activity and the integrity of the structure.

Unless otherwise shown on the drawings or directed by the Employer, the minimum stripping times shown in Table **4-1**, shall be observed for the various sections of the works:

Table 4-1 : Minimum Formwork Stripping Times

Section of Formwork Removed	Minimum Stripping Time (Days)
Beam sides, walls and unloaded columns	3
Slabs with props left undisturbed underneath	7
Beam soffits with props left undisturbed underneath	7
Removal of props under slabs	14
Removal of props under beams	14

In the event that the ambient average temperature falls below 10°Centigrade or the minimum falls below 5°Centigrade over a 24 hour period, all minimum stripping times shall be increased by 100%.

Where not included in this table, the minimum stripping times shall be as recommended in ACI PRC-347.

After removal of formwork, concrete surfaces shall be inspected by Employer for defects. Any concrete surface which has had remedial work carried out on it, prior to being inspected, shall be rejected.

If inspection of concrete surfaces reveal small holes and minor honeycombing (under 20 mm x 20 mm), which do not impair strength, they shall be filled with mortar. Mortar shall be of essentially the same composition as the surrounding concrete. Tie rod holes and honeycombing with size greater than (20 mm x 20 mm) and members with defects (including but not limited to bulging, warping etc) shall be repaired or replaced.

The Work Method statement, for the repair works to be used in the removal and replacement of defective work, shall be submitted to the Employer for approval before the removal commences. Special care shall be given to the joint between the original and repair concrete.

If the Employer requires the defective concrete to be removed and replaced and/or is not satisfied with the repair works, the defective concrete shall be removed and replaced at no additional cost to the Employer.

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4.5 Tolerances

Unless otherwise specified on the drawings tolerance for finished concrete dimensions shall be in accordance with ACI SPEC-117 – Standard Specification for Tolerances for Concrete Construction and Materials. No tolerance shall apply to the minimum specified cover to reinforcement.

4.6 Holes, Grooves and Embedded Items

4.6.1 *Formation and Fixing*

All holes, grooves, chases and nibs shall be formed during the casting operation, unless otherwise approved by the Employer.

Anchor bolts, pipe sleeves and other embedded items shall be accurately set and securely held in position. Immediately after placement of the concrete, the Contractor shall check the position of all embedded items and advise the Employer of any errors detected.

Chamfer strips shall be used for all exposed exterior corners, to produce 20 mm beveled edges.

4.6.2 *Location*

If the position of any of the above embedded items is found to interfere with major reinforcement, structural member or piece of equipment, the Employer shall be informed immediately. The Employer shall direct the Contractor in writing regarding remedial measures to be taken.

4.6.3 *Anchor Bolts*

Anchor bolts shall be standard ISO metric hexagon commercial bolts.

Anchor bolts shall be fabricated from plain round steel bar conforming to MD-101-SMEM-EG-CV-SPC-0002, Anchor Bolts or as nominated on the drawings. All steel shall be clean, free of rust, scale, oil and any other detrimental matter. Hooks and cogs shall be cold formed.

4.6.4 *Anchor Bolt Placement*

A solid timber or steel template shall be used to maintain the exact position of the bolts during casting operations, and such templates shall be rigidly fixed to the formwork.

Where bolts are detailed on the drawings with base plate and sleeves, full length sleeves of approved material shall be fitted centrally around each holding down bolt and shall be kept in firm contact with the base plate.

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On completion of casting operations, the templates shall be removed, taking care not to disturb the position of the bolts. Any concrete surfaces obscured by the templates shall then be finished as required.

Where shown on the drawings, other fixtures shall be set in the concrete to the details given and to the approval of the Employer.

Embedment shall be installed to the tolerances shown in Table 4-2, unless noted otherwise on the drawings.

Table 4-2 : Embedment Tolerances

Anchor bolt projection	+6 mm, –0 mm
Centre to centre of any two anchor bolts within a group	+ or – 2 mm
Centre to centre between bolt groups	+ or – 4 mm
Anchor bolt plumbness	4 mm in 1 m
Plate inserts	+ or – 6 mm horizontal and + or – 1 mm vertical

4.7 Jointing of Concrete

4.7.1 *Control Joints*

All control joints shall be in the positions and formed in accordance with the details as shown on the drawings and the design of the joints shall conform to ACI PRC-302.1 and ACI PRC-224.3.

Recesses for control joint sealants shall be accurately formed and filled with the sealant nominated on the drawings. Sealing compound and expansion joint material shall be installed in strict accordance with the Manufacturer recommendations, a copy of which shall be submitted to the Employer for approval prior to the commencement of concrete joint work. If curing compounds have been used on the concrete, any compound film on concrete faces where a sealant bond is required shall be cleaned off by a light whip blast with a garnet agent. This removal shall be to the satisfaction of the Employer and shall provide a clean fresh concrete surface for contact with the sealant compounds. Care shall be taken that no damage occurs to any cast-in joint filler during the cleaning process.

Control joints to be cut with a saw shall be cut as soon as the concrete is hard enough to prevent surface raveling and aggregate dislodging and no later than 12 hours after concrete placement. Concrete joints shall be cut in accordance with the saw Manufacturer written recommendations. Sawing sequence shall be based on pour time and size of slab.

4.7.1.1. Isolation Joints

Isolation joints shall be placed where the pavement adjoins vertical surface (e.g. walls, columns, catch basins, manholes and equipment foundations).

4.7.1.2. Expansion Joints

Expansion joints shall be constructed with preformed expansion joint filler material or another approved device to allow for expansion. Joint sealer shall be provided. Dowels at expansion joints shall be properly aligned to prevent any restraint on expansion movement at the joint. The expansion joints for slabs on grade, wall construction, and other appropriate structures, shall be provided on centers up to a maximum of 10m, unless otherwise stated in the Drawings. These joints shall be constructed so that they pass completely through the concrete section. Expansion joints shall extend till the complete section of the slab. Pre-moulded expansion joints material shall conform to ASTM D1751.Joint sealant materials shall conform to ASTM D6690, ASTM D5893/D5893M for horizontal joints, or ASTM D2628 for both vertical and horizontal joints.

4.7.1.3. Contraction Joints

The contraction joints for slabs on grade, wall construction and other appropriate structures shall be provided on centers up to a maximum of 5 m, unless otherwise stated in the Drawings. These joints shall be formed by scoring of the concrete prior to taking its initial set or saw cutting within 12 hours after placing. They shall extend a minimum of one-fourth of the thickness of the concrete section and shall follow requirements of ACI PRC-224.3.

4.7.2 *Construction Joints*

All construction joints shall be in the positions and formed in accordance with the details as shown on the drawings and the design of the joints shall conform to ACI PRC-302.1 and ACI PRC-224.3.

The construction shall be monolithic to the maximum extent possible. All joints shall be located and constructed in accordance with the construction drawings. Where additional construction joints are required by the Contractor for any reason, the Contractor shall obtain written approval for his preferred locations from the Employer prior to pouring the affected portion of the works, and all costs for the additional joints shall be borne by the Contractor.

Construction joints shall be at right angles to the main reinforcing bars in the member, unless otherwise shown on the construction drawings. Stop ends for vertical joints shall be removed as soon as it is practical without damaging the concrete face. Construction joints shall be formed by inserting a temporary vertical stopping board against which concrete can be properly consolidated. At the discretion of the Employer, an arris shall be formed along the exposed corner to ensure a straight joint on the finished surface.

The exposed faces of all previous work where a construction joint has been formed and against which new concrete is to be placed shall be properly cleaned and roughened within 12 hours of casting by air or water or wire brushing or scrabbling to show the tips of the coarse aggregate.

If the set has progressed too far and this is not practicable, the previous work shall be bush hammered and cleaned to present a suitable surface against which new work may be cast. Reinforcement shall continue through construction joints.

Immediately prior to casting new concrete against existing concrete, the exposed face of the existing concrete shall be primed with an approved bonding agent prior to placement of new concrete. Concrete joint surfaces to receive new concrete shall be saturated, but free of surface water.

4.7.3 *Water Stoppers*

Water stoppers shall be installed as indicated in the Drawings and in accordance with the Manufacturer recommendations and ACI PRC-224.3.

The water stoppers shall be accurately cut, fitted and fully joined by welding at joints and junctions (Per water stop Manufacturer recommendations) to provide a continuous, watertight diaphragm at all points.

The lapping of water stops at joints, and use of adhesive for jointing purposes, shall not be permitted unless approved by Employer. Joints in water stops shall be tested for water tightness before pouring of concrete.

Adequate provisions shall be made for the support and protection of water stoppers during the progress of the work. Care shall be taken at all times to ensure that water stops are not perforated or damaged in any way during concrete placement. Damaged water stops shall be replaced or repaired before embedment in concrete. Particular attention shall be given to placement and vibration of the concrete to ensure well compacted concrete around the ribs of the water stops. Damaged stoppers shall be replaced and/or repaired before embedment in concrete.

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4.8 Placement of Concrete

4.8.1 *Labor and Equipment*

No concrete placement shall commence until all necessary equipment for placement, compaction and finishing are at the location of the pour and in good working order, and inspection of formwork and reinforcement has been conducted and approved by Employer. Arrangements shall be made to support all walkways and barrow runs from the blinding concrete or shuttering, clear of the reinforcement and service ducts. All operations shall be carried out without imposing any load on the reinforcement, formwork, false-work, or any embedded item that might cause its distortion or dislocation from its intended position.

The Contractor shall make available an adequately sized and competent labor force for the operation at hand. During concreting, competent steel fixers shall be in attendance to adjust and correct the position of the reinforcement if necessary. Competent carpenters shall also be present to ensure all formwork is maintained in place so as to conform to the requirements of Section 4.4 of this specification.

Equipment for chuting, pumping and conveying concrete shall be of such size and design as to ensure a continuous supply of concrete at the point of delivery without segregation of the materials. All such equipment shall be subjected to the Employer's approval and shall not be used without his consent. Concrete delivery equipment shall conform to ACI SPEC-301 and ACI PRC-304.

4.8.2 *Placement*

No concreting shall ever be done against soil.

Care shall be exercised to ensure that concrete already placed is not damaged or otherwise disturbed by further operations in adjacent areas. The concrete shall be placed in a planned manner continuously between construction joints, generally progressing in one direction, without making any cold joints either horizontal or vertical.

Where the concrete surface is inclined, work shall progress in the rising direction unless otherwise approved by the Employer or shown on the drawings.

Concrete shall not be dropped through heights exceeding 1 m, but transferred by Tremie pipe and chute method, or other methods approved by the Employer.

All concrete shall be thoroughly consolidated until it has penetrated and filled all spaces in the forms in such a manner as to form a monolithic mass, properly bonded to any previously placed concrete.

Concrete shall be placed in layers based on the requirements stated in ACI SPEC-301 and ACI PRC-304, each layer being shallow enough to ensure that it can be fully vibrated, including penetration into the previous layer.

The concrete shall be finished accurately to the levels, shapes and sizes shown on the drawings and in accordance with the tolerances required by this specification.

4.8.3 *Hot or Cold Weather Concreting*

The Contractor shall submit for the Employer's approval, details of proposed arrangements for the placement of any concrete where the temperature of the air surrounding the concrete pour is in excess of 32°C or is below 5°C.

The temperature of the concrete shall be within the range of 10°C to 32°C, at the point of delivery, unless otherwise agreed with or requested by the Employer in writing.

Concrete temperatures shall be recorded at the time of placement, as directed by Employer. If concrete is required to be placed in ambient temperatures below 5°C, adequate protection of the concrete after placement shall be provided by covering with insulation. For placement of concrete in ambient temperatures above 32°C, the Contractor shall ensure that the holding time in the transit mixer is kept to a minimum. When the ambient temperature exceeds 32°C, the concrete placement shall be in accordance with ACI SPEC-305.1. The temperature of fresh concrete shall be maintained between 25°C and 32°C, and at time of placement shall not exceed 32°C. Any concrete with a temperature in excess of 32°C at the time of placement shall not be used.

Placement of concrete during hot weather shall comply with Section 5.13, SBC 304, and ACI SPEC-305.1

Windbreaks shall be placed prior to placing concrete when the prevailing wind exceeds or is expected to exceed 15 km/hr within 24 hours of concrete placement. Windbreaks shall be positioned to minimize the rapid drying of exposed concrete surfaces and shall be maintained in place for a minimum of 24 hours following concrete finishing.

An approved, pigmented anti-evaporative agent shall be applied to all concrete surfaces immediately after scratching while surface moisture is still present. Application shall be strictly in accordance with the Manufacturer instructions. Under severe drying conditions the Employer may direct the application of additional anti-evaporative agent after each finishing operation.

Specific requirements for hot weather curing are outlined in Section 4.9.2 of this specification.

The Contractor shall advise the Employer regarding delivery of concrete from distant batch plants. Any ready-mixed concrete not placed within 60 minutes of adding water may be rejected by the Employer. Concrete delivery shall be constant during a scheduled pour.

Any interruptions or undue delays in the delivery of concrete may result in rejection by the Employer of any part of the concrete, and it's removal and rectification shall be carried out at the Contractor's expense.

4.8.4 *Compaction of Concrete*

All structural concrete shall be consolidated by vibration to a plan and in a manner approved by the Employer.

For concrete work, excluding slabs of thickness less than 150 mm, poker vibrators of appropriate size and of adequate number and power shall be used, in accordance with ACI PRC-309. Slabs of thickness less than 150 mm shall be compacted by means of approved vibrating screeds.

Equipment used for consolidation shall be kept in good working order at all times and there shall be adequate standby units available.

4.9 Curing of Concrete

4.9.1 *General*

Concrete shall be moist cured. Curing shall be commence immediately after the finishing operation or within two (2) hours after concrete setting, and shall continue for a minimum of 14 days.

All structural concrete shall be cured in accordance with the requirements of ACI SPEC-301, ACI SPEC-308.1 and ACI SPEC-305.1.

For concrete pours using ordinary Portland cement, provided no additional concrete or finishes are to be applied to the cured concrete, curing shall be by spraying all surfaces with curing compound directly after scratching.

For all concrete pours using low heat cement thermal curing shall be carried out and ensured that the maximum temperature differential across the concrete element does not exceed 20°C, or the maximum core temperature does not exceed 60°C at any time. Where thermal curing is specified on the design drawings for concrete pours using ordinary Portland cement, the differential and maximum temperature limits shall be agreed between the Contractor and the Employer. The temperature gradient in slabs and walls over 600 mm thick shall be verified by computer models and trail mix test pieces. If the temperature gradient is more than 20° C, adequate measures shall

be taken to reduce the heat of hydration (e.g. by cooling, by using insulated formwork, and/ or by applying low-heat cement, retarder, pozzolanic material, etc.)

The Contractor shall submit the proposed thermal curing procedures to the Employer for approval at least seven days prior to commencing such pours. For concrete pours requiring thermal curing, the Contractor shall measure and record concrete temperatures at locations in the concrete block agreed with the Employer to confirm compliance with the temperature limits specified.

Only curing methods approved by the Employer shall be used. Slabs shall be covered with a membrane after finishing. Curing compounds shall comply with ASTM C494/C494M and be approved by the Employer. Curing compounds shall be applied in accordance with the Manufacturer instructions.

Where additional concrete or a non-compatible finish is required to be applied to the cured concrete surface, water curing shall be used, or surfaces shall be wrapped with impervious plastic membrane. However, in the latter case the enclosure shall be completely sealed on all sides by plastic welding or by clamping along the edges.

4.9.2 *Hot Weather*

No curing compound shall be used when the ambient temperature exceeds 32°C.

Vertical surfaces shall be protected by impermeable shutters and shaded from the direct heat of the sun. The Contractor shall carry out curing at initial set by fog sprays and commence water curing as soon as is practicable after finishing operations are complete and continue for 14 days.

4.10 Finishes

4.10.1 *Concrete Floor Surfaces*

The characteristics of concrete floor surface finishes shall be as directed on the drawings and in accordance with the classification system of ACI PRC-302.1.

4.10.2 *Unformed Surfaces*

All exposed corners including top of walls shall be chamfered. Upper surface finishes shall be in accordance with the following classes:

a) Scratched

This is a scratched finish intended for the surfaces of roads or of foundations and structural members to be covered by backfill or subsequent stages of construction. The finishing operations

shall consist of leveling and scratching the concrete to produce a uniform, plain or ridged surface, surplus concrete being struck off by a straight edge immediately after consolidation.

b) Wood Float

This is a floated finish for exposed surfaces where a hard smooth steel troweled surface is not required. Floating shall be carried out after the concrete has hardened sufficiently and may be by hand or machine. Care shall be taken that the concrete is worked no more than necessary to produce a uniform surface free from screed marks.

c) Steel Trowel

This is a hard smooth steel troweled finish for tops of walls, copings and other members exposed to weathering, surfaces of foundations, and seatings for bearing plates and the like where the metal is in direct contact with concrete. Troweling shall not commence until the moisture film has disappeared and the concrete has hardened sufficiently to prevent excess laitance from being worked to the surface. The surfaces shall be troweled under firm pressure and left free from trowel marks.

d) Broomed

This is a brushed finish for surfaces of pavements. It shall be finished as specified for smooth steel troweled finish, except that the hard steel troweling shall be omitted. After the machine troweling is completed, the surface shall be brushed with a fiber bristle brush or broom across the direction of traffic. Hardeners or other surface treatments as specified on the drawings shall be applied to surfaces where indicated, in accordance with the Manufacturer instructions.

5. PRECAST CONCRETE

5.1 Materials

Concrete and reinforcement shall comply with the requirements of ACI PRC-523.2, ACI PRC-533, ACI CODE-318 and ACI ITG-7. The concrete strengths and reinforcement properties shall be as noted on the Drawings and/or as specified elsewhere in this specification.

Jointing material to precast panels shall have details as indicated on the Drawings.

Provide sealants and fire rating material in accordance with the Manufacturers published recommendations. Concrete admixtures, where they are specifically approved in writing by the Employer, shall be in accordance with ACI CODE-318.

Use of silica fume is not permitted for precast wall panels and slabs on grade (building floor and paving)

Lifting attachments such as ferrules or other types of cast-in fittings shall be recessed to allow patching with an epoxy mortar. Cast-in assemblies, fittings and fixings which will be exposed before or after erection shall be hot dipped heavy duty galvanized after all fabrication operations of them have been completed.

If tilt-up techniques are to be used for pre-casting the concrete curing compound and release agent (bond breaker) shall be compatible. A proven proprietary combined curing compound/release action which will allow the specified finishes to adhere to the surface shall be adopted and shall be applied in accordance with the Manufacturer specification.

5.2 Connection, Inserts and Holes

The locations, sizes and types of all fixing, hoisting, handling and erection devices shall be described on the Drawings.

Lifting ferrules on the panels shall be marked distinctively so as to be easily identifiable from other types of ferrules. All components shall be located accurately and the locations shall not be altered.

All bolts, nuts, washers, lifting inserts and ferrules shall be hot dipped galvanized.

Leveling pads shall be provided consisting of Compressed Fiber Cement shims, In-situ Concrete, or Rigid High-Impact PVC shims. Concrete leveling pads shall be at least 500 mm long and the full thickness of the panel. PVC shims should be at least 150 mm long.

5.3 Identification

Each precast unit shall be clearly and permanently identified, as shown on the Shop Drawings. The identifying mark is placed on a surface of no visual significance and should include the date of casting. The same number shall not be given to more than one unit.

5.4 Design

The reinforcement shown on the Contract Drawings is mandatory and has been determined for in-service loads only. The reinforcement shall be assumed to be a minimum but may be increased, any modifications being entirely at the Contractor's expense. The Contractor is responsible for designing and providing any additional reinforcement and/or strong backs that may be necessary to suit his lifting and/or erection procedure.

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The Contractor shall engage a qualified structural engineer to certify the shop drawings, strong backs (where required) and erection procedure, including the bracing of the panels. Written evidence of this certification shall be provided to the Employer.

5.5 Curing

The production schedule shall allow for curing precast concrete in accordance with this Specification.

5.6 Preparation For Lifting

Slinging, transporting, stacking and installation may take place when the concrete strength designed for theses purpose has been attained.

Bracing shall be prepared for immediate connection after the panels are lifted.

The Contractor shall be responsible for the safety of operations at all stages of the lifting and handling of the panels.

5.7 Tolerances

Tolerances on the indicated finished dimensions of the precast panels at the time of assembly in the building shall be in accordance with ACI ITG-7.

5.8 Defective Panels

Panels may be rejected by the Employer if they do not meet specification requirements or if they are damaged during manufacture or erection.

The standard of finish and general appearance of the panels must be of a standard commensurate with its use, location and final surface coating. Rough finishes of surfaces, edges and arises will not be tolerated.

Any rejected panel shall be replaced with another panel at the Contractor's expense.

5.9 Lifting and Handling

The Contractor shall be responsible for the erection and method of erection, design and installation of bracing and shoring of the panels during construction, until the structure has been sufficiently completed to enable the panels to carry their own loads.

Precast panels shall be handled, lifted and positioned in a manner which does not cause damage or cracking. The weight of panels shall be taken up gradually and without snatch. When panels are being lowered, they shall be let down gently into position without impact.

Panels shall be adequately braced and supported during handling and erected and temporarily braced to ensure proper alignment and safety.

The panels shall be lowered on to the leveling pads previously set at pre-determined levels. Each panel shall be located such that no alterations to the fixing are made, and that no subsequent slope or movement occurs, and that no undesirable stresses are built into the panels or connections.

All necessary frames, braces, guys, wedges and other supports required to secure the panels in place shall be provided.

All grout, temporary fixings, shims, braces, mortar, fireproofing, protection, sealant, joining strips and flashings required for erection of the precast panels on the site shall be installed.

If in the opinion of the Employer, any panel after erection does not conform to the conditions specified for appearance and position, it shall be taken down and replaced at the cost of the Contractor.

5.10 Joint Sealing

Completing of joint sealing shall be carried out only after the completion of installation of precast panels.

All joint sealing shall be installed by an approved specialist sub-contractor who is appropriately equipped for the work, skilled in the sealing of joints, and recommended by the Manufacturer and/or Supplier of the sealant as an approved applicator.

5.11 Protection

Precast panels shall be protected from damage from the time of removal from the forms until the completion of the project. In particular, panels shall be protected against damage from the local crushing and chafing effects of lifting and handling equipment.

Provide additional protection required where concrete will be poured at levels above units which have been erected.

All vulnerable surfaces, edges, corners, etc. shall be protected for the duration of the contract. Materials and methods shall be adopted which will not harm precast materials.

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5.12 Cleaning

All visible panel surfaces shall be thoroughly cleaned, and a uniform finish shall be achieved throughout. The completed work shall be inspected by the Employer, the Contractor advising of appropriate timing for the inspection.

6. LIQUID RETAINING STRUCTURES

6.1 General

Liquid retaining structures material and design shall comply with ACI CODE-350. Where the water to be retained is saline or corrosive, evaluate use of cathodic protection or use of GFRP reinforcement. The internal face of the structures shall be coated, based on the coating Manufacturer's recommendation.

6.2 Joints

Joints shall be designed and constructed in accordance with ACI PRC-224.3 Chapter 34 ASTM 5893 and ASTM 6690.Where water-stops are required and/or are indicated on the drawings for joints, they shall be fabricated from PVC in accordance with CE CRD-C572. Jointing of water-stops shall be by welding in accordance with Manufacturer recommendation. The lapping of water-stops at joints and the use of adhesive for joining purposes, shall not be allowed unless approved by Employer. Care shall be taken at all time to ensure that water-stops are not perforated or damaged during concrete placement. Particular attention shall be paid to the placing and vibration of the concrete to ensure well consolidated concrete around the ribs of water-stops. At all joints, the concrete shall be placed up to the center line of the water-stops.

Wall starters to walls of the watertight construction shall be cast using formworks, to allow the concrete in the wall starters to be placed simultaneously with the concrete in the slab.

6.3 Testing

Tests to determine the overall water tightness of the structure and the joints shall be carried out per ACI CODE-350. The structures under test shall be filled to the designed level with potable water and left for a 7-days period to allow for absorption of water. The exposed faces shall be inspected for leaks over this period. Any defect shall be corrected by an Employer approved method. This could involve demolition and rebuilding or lining of the structure in case of leaks at the joints.

7. PROTECTIVE COATING FOR CONCRETE

Concrete surface shall be protected as specified in MD-101-SMEM-EG-ST-SPC-0008, Protective Coating for Concrete.

8. INSPECTION AND TESTING

The contractor shall comply with the Project Quality Requirements for specific QA/QC requirements.

8.1 Records

The Contractor shall keep accurate, up to date records of the concrete work as listed below. Marked up drawings and all test results shall be retained on site until the completion of the Contract:

- Details of sections of work as concreted shall be marked up on the General Arrangement drawings with construction joints, pour date and relevant test cylinder reference numbers
- Quantity and class of concrete for each section poured shall be noted, and
- Concrete test results, obtained in accordance with Section 8.3 of this specification, including compressive strength at seven days and 28 days and slump measurement shall be retained on site. Copies of all test certificates shall be supplied to the Employer within seven days of the test being completed.

8.2 Inspection and Approval

All works, stages of the works, and materials and processes, shall be made available for the inspection and approval by the Employer.

The Contractor shall give a minimum of 24 hour's notice to the Employer and obtain his written consent before placing any concrete.

A Request For Inspection (RFI) shall be submitted by the contractor requesting for inspection and seeking approval of major activities including but not limited to site preparation, setting out (survey), lean concreting, form work/shuttering, steel reinforcement work. The contractor shall proceed for subsequent activities only on approval from the Employer.

8.3 Concrete

The Contractor shall be responsible for concrete sampling and testing in accordance with ACI SPEC-301 and MD-101-SMEM-EG-ST-SPC-0005, Premixed Concrete, at the frequencies specified in the Table 8-1 below. Each sample set shall consist of three test cylinders (2 of test cylinders shall be cured under standard moisture & temperature condition per ASTM C31 and one cylinder (spare)

shall be cured in Field for a minimum period of 28 days under conditions similar to main concrete) which shall be tested by an Employer approved laboratory as follows:

- 1 at 7 days after placement of the concrete
- 1 at 28 days after placement of the concrete (56 days in the case of low heat cement concrete)
- 1 spare. To be tested when directed by Employer
- If the test results indicate non- compliance, the quantity of concrete represents by the results shall be removed and replaced with concrete meeting the requirement of this specification at no additional expense to Employer. Additionally, an investigation shall be performed by Contractor to identify the cause of non- compliance and specify corrective measures. The extent of the investigation, and contractor's proposed corrective measures, shall be subject to Employer's approval.

Table 8-1 : Concrete Sampling Frequencies

Pour Size in ($m^{3)}$	Number of Samples
1 to 5	1
6 to 25	2 (1st batch and 1 other)
26 to 50	3 (1st batch and 2 others at random)
51 to 100	4 (1st batch and 3 others at random)

For each additional 50 m3 of any pour, one (1) additional sample set shall be taken. The Contractor shall adequately protect the test cylinders during their setting and curing period on Site and during handling and transport to the test laboratory.

8.4 Rejected Concrete

All concrete work rejected in accordance with this specification and concrete rejected in accordance with ACI SPEC-301 shall be removed where directed by the Employer. Non-destructive testing of concrete including impact or rebound hammer and ultrasonic methods or any combination of these shall not be acceptable methods of testing. The Employer may ask the contractor to carry out testing in these methods, but still the Employer has the right to reject the concrete with or without doing such test, and results of such tests may or may not be used by the Employer for reconsideration. The decision of the Employer is final, requiring no justification.

Where hardened concrete is rejected, the adjoining concrete shall be cut back as directed by the Employer. The rejected concrete work shall be reconstructed to the requirements of this Specification and the specific directions of the Employer. The adjoining concrete cut back during the rejection and reconstruction shall be repaired to the approval of the Employer.

All rejected concrete, including adjoining concrete identified to be part of the repair, shall be replaced at the Contractor's expenses. The removal and replacement of all non-complying material shall not be the basis for any claim for any additional costs or extension of time to the schedule.